



SIMATS ENGINEERING
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Course Code: CSA0509

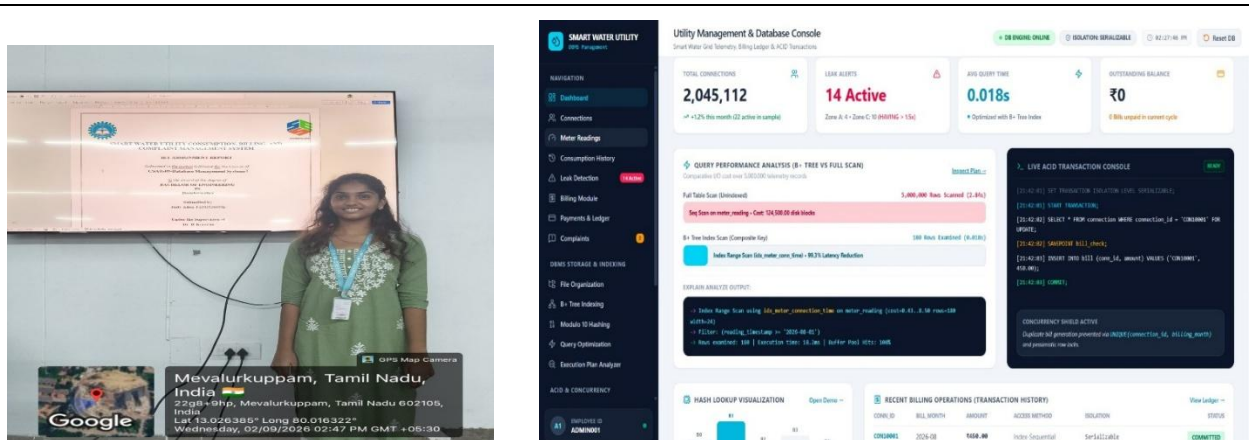
Slot: A

Course Name: DATABASE MANAGEMENT SYSTEM

Course Faculty: Dr. R Kesavan

Project Title: Smart Water Utility Consumption, Billing, and Complaint Management System

Module Photographs: Smart Meter Data Storage and Retrieval



Introduction:

This module manages the storage and retrieval of smart water-meter readings. Indexed-sequential organization, B+ tree indexing, and hashing are used to reduce retrieval time and avoid full-file scans.

Objectives:

- Store hourly meter readings efficiently.
- Retrieve consumption history quickly.
- Retrieve the latest reading using hashing and detects unusually high consumption.

Methodology:

Meter readings are stored with Connection ID, Zone ID, timestamp, and consumption value. A composite index on Connection ID and timestamp supports history queries, while hashing is used for latest-reading lookup.

Conclusion:

The combination of indexing and hashing provides a fast and scalable solution for smart-meter data retrieval.

Student Signature

Guide Signature